VideoDetective: Clue Hunting via both Extrinsic Query and Intrinsic Relevance for Long Video Understanding

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Research Questions

- How can we effectively localize query-relevant segments in long videos while leveraging both extrinsic query information and intrinsic video structure?
- How can sparse observations of video segments be propagated to estimate global relevance across an entire long video?
- Can a graph-based belief propagation framework improve long video understanding across different MLLM architectures?

Core Idea

Long video understanding is challenging because MLLMs have limited context windows, requiring selective sampling of relevant segments. Existing methods rely purely on query-to-segment matching, ignoring video structure. VideoDetective's key insight is that videos have intrinsic temporal and visual correlations - observing one segment provides information about semantically or temporally related segments. The method models videos as visual-temporal affinity graphs and uses iterative belief propagation to estimate global segment relevance from sparse observations, enabling "see less but know more" through graph diffusion of relevance scores.

Key Formula

Iterative Belief Diffusion

$$F^{(t+1)} = \beta W_{norm} F^{(t)} + (1-\beta) Y^{(t+1)}$$

F^(t) is the belief field (global relevance scores), Y^(t) is the injection vector (sparse observations), W_norm is the normalized affinity matrix, β balances smoothness vs consistency. This propagates sparse relevance scores across the graph to update global beliefs about segment relevance.

Multi-Route Prior Score

$$(Y_r^{prior})_i = \alpha \cdot \max_{w \in K_r} \langle \phi_T(w), h_i \rangle + (1-\alpha) \cdot \max_{p \in P_r} \langle \psi(p), \psi(e_i) \rangle$$

Y_r^prior is the prior relevance for facet r, K_r are keywords, P_r are semantic descriptions, φ_T is text encoder, ψ is semantic encoder, h_i is segment feature, e_i is event description. This fuses sparse visual matching (keywords) and dense semantic matching (descriptions) for initialization.

Key Figure

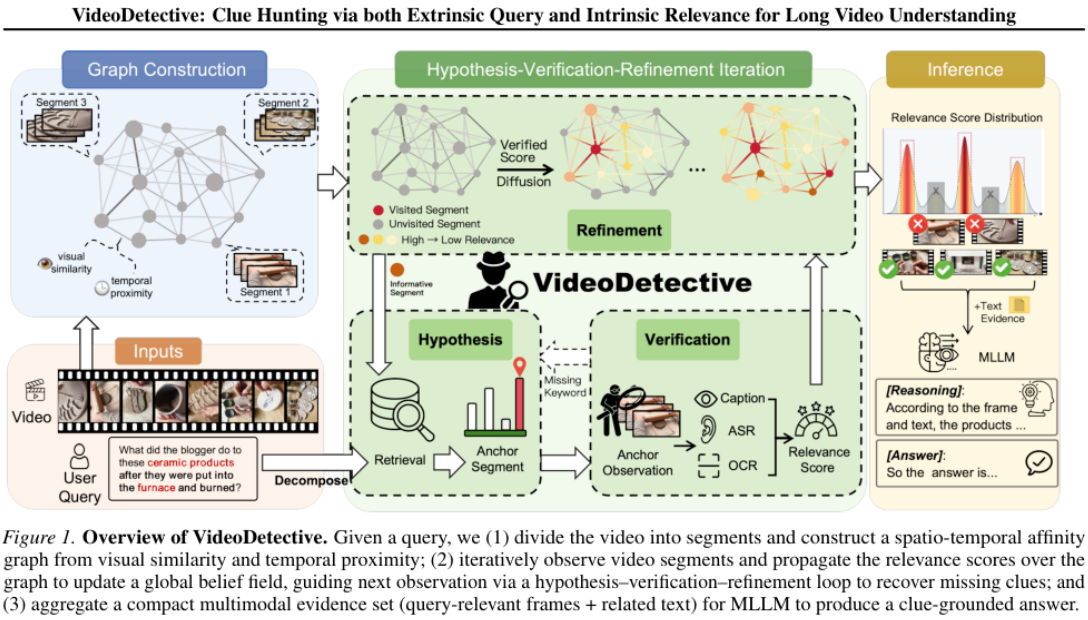

Figure 1. **Overview of VideoDetective.** Given a query, we (1) divide the video into segments and construct a spatio-temporal affinity graph from visual similarity and temporal proximity; (2) iteratively observe video segments and propagate the relevance scores over the graph to update a global belief field, guiding next observation via a hypothesis–verification–refinement loop to recover missing clues; and (3) aggregate a compact multimodal evidence set (query-relevant frames + related text) for MLLM to produce a clue-grounded answer.

Figure 1: Shows the complete VideoDetective pipeline including graph construction, hypothesis-verification-refinement loop, and evidence aggregation for final MLLM answering.

Limitations

- Relies on VLM self-reflection capability to provide feedback signals (e.g., 'missing keywords'), which may be unreliable
- Graph construction depends on visual similarity thresholds and temporal decay parameters that may need tuning
- Computational overhead of graph construction and iterative diffusion may limit scalability to very long videos

- Performance still bottlenecked by the visual encoder capability rather than reasoning capacity